

DERBY VALUE

Smart Ticket Scanner · Backtest & Process Reference

Churchill Downs · Oaks Day May 1, 2026 & Derby Day May 2, 2026

PART 1 — BACKTEST RESULTS

Two Churchill Downs race days were analyzed across 27 races total. The backtest validated the core signals used in the scanner — the oNg gap signal, EV edge model, and BOMB tag — and revealed two distinct exotic patterns that shape how tickets are constructed.

Oaks Day — Friday, May 1, 2026

Sequences that hit	P4 R10-11-12-13	P5 R9-10-11-12-13	P6 R8-9-10-11-12-13
3	\$1,221	\$9,404	\$43,529

Stakes race winners — oNg gap signal

Race	Winner	Race	Gap	Signal	Notes
R8	On Time Girl	Eight Belles G2	+10	VALUE	P6 start anchor. Signal caught.
R9	Corporate Power	Alysheba G2	+3	BOMB	Gap below threshold. BOMB tag was the tell.
R10	Kathynmarissa	Modesty G3 · 24-1	+2	BOMB	24-1 winner. BOMB signal overrode gap miss.
R11	Shred the Gnar	La Troienne G1	+7	FADE	Gap/FADE conflict — broke all three chains.
R12	Imaginationthelady	Edgewood G2	+3	—	Below threshold. Top gap horses did not win.
R13	Always a Runner	Kentucky Oaks G1	+7	KEY	Anchor of all 3 sequences. \$13.04 win.

Pattern — Bookend: The gap signal caught R8 (On Time Girl, +10) and R13 (Always a Runner, +7) — the start and end of all three winning sequences. The four middle races were opaque. The R11 FADE conflict (Shred the Gnar: gap +7 but FADE-tagged) was the critical break point — the scanner's old rule excluded it entirely, which would have broken every chain.

Derby Day — Saturday, May 2, 2026

OptixEQ key hit rate	P3 R12-13-14	Ticket cost	EV FADE calls correct
7 / 14	~\$700-1,100	\$2	3 / 4

P3 R12–13–14 — consecutive gap signal · all three winners

Race	Winner	Race / Payout	Gap	Signal	Notes
R12	Golden Tempo	Kentucky Derby G1 · \$48.24	+16	VALUE	Single. oFIG 88 → oNg 104. Won at 23-1.
R13	Gilded Bandit	AOC · \$5.68	+4	—	Marginal catch at threshold ≥4.
R14	Prize Pick	MSW 2YO · \$15.88	+14	—	Single. Turf→Dirt debut. Structural underbet.

Pattern — Consecutive: Three adjacent races all had oNg gap winners. All three singled. A \$2 straight P3 ticket. No FADE conflicts. Numeric oNg (Derby Day) made the signal far cleaner than the letter-grade oNg used on Friday.

Gap signal wins: 2 of 14 races	Gap signal wins: 2 of 7 stakes
Pattern: Consecutive — R12, R13, R14	Pattern: Bookend — R8 and R13 only
FADE conflicts: none	FADE conflict: R11 Shred the Gnar broke chains
Ticket: 3 singles = \$2 P3	Ticket needed: wide in 4 middle legs
oNg type: numeric — precise	oNg type: letter grade — noisier
Est. P3 payout: ~\$700–\$1,100	P6 payout (if on): \$43,529 but ~\$48 ticket

PART 2 — SMART TICKET SCANNER PROCESS

The scanner runs automatically when the Today tab is opened. It scans every P3, P4, and P5 window on the current race card and returns up to 6 ranked tickets. Each ticket shows the horses selected per leg, why each was chosen, and an upgrade hint when a P3 can be cheaply extended to a P4.

1

Position selection

Filters every P3/P4/P5 window at the chosen track by historical overlay ratio (actual exotic payout ÷ theoretical parlay value). Only windows at or above the minimum threshold (default 1.0×) proceed. The CD P4 R4-5-6-7 at 2.56× overlay always surfaces first. Early-card P3 windows below 0.5× are skipped.

2

Horse enrichment — stack all signals per leg

For every horse in each leg: the EV model computes model win% vs market ML-implied%, calculates the oNg gap (projected next-race figure minus past figure), and reads OptixEQ tags. FADE horses with gap ≥6 are kept as secondary candidates. BOMB-tagged horses at 15-1 or longer are added to the candidate pool regardless of gap threshold.

3

Selection tier — single, pair, or spread

Each leg independently assigned. SINGLE if model% ≥35% or gap ≥8 with KEY/VALUE — but never if FADE-tagged. PAIR if top-two combined ≥52% or gap ≥4. SPREAD (3 horses) for uncertain or FADE-conflict legs.

4

Bookend detection

After all legs are scored: if a 4+ leg sequence has strong signals only at the first and last legs with no strong middle signals, middle legs are auto-widened to 3 horses and labeled BOOKEND. Derived from Oaks Day — On Time Girl (R8, gap+10) and Always a Runner (R13, gap+7) caught the endpoints; 4 races between them were opaque.

5

Cost filter

Total combinations × \$2 base must not exceed the max cost setting (default \$20). Bookend expansion can push tickets over budget — raise max cost to \$50 if you want bookend coverage on P5/P6 sequences.

6

Ticket scoring and ranking

Score = overlay ratio + EV edge bonus + gap signal bonus + strong-signal bonus, penalized by combination count (wider = lower score). Top 6 tickets returned in order. The top-scoring ticket gets the TOP PICK label.

7

P3 → P4 upgrade hint

For every P3 ticket: checks if a P4 extension exists where the 4th leg can be singled (model% ≥30% or gap ≥6). If so, the upgrade is shown with estimated payout multiplier. At CD the median P3→P4 multiplier is 4.6×. Adding a singled 4th leg keeps cost the same while multiplying the expected payout.

Selection tier rules

SINGLE (1 horse)	PAIR (2 horses)	SPREAD (3 horses)
Model% ≥35% or gap ≥8 + KEY/VALUE tag or gap ≥8 + model% ≥20% Never if FADE-tagged.	Top-2 combined ≥52% or gap ≥4 (not FADE-tagged) FADE/gap horses may appear here as secondary.	Open or uncertain field or bookend middle leg or FADE horse is top of field

Signal definitions

GAP +N

Projected next-race figure (oNg) minus past figure (oFIG). Positive = OptixEQ projects improvement the market has not yet priced in. Numeric oNg (from Derby Day data) is precise. Letter-grade oNg (from Oaks Day data) is noisier — more horses share the same grade band.

Rule: Thresholds: ≥ 8 = single candidate. ≥ 4 = include in exotics. ≥ 6 overrides FADE exclusion.

Example: Golden Tempo: oFIG 88 $\rightarrow$ oNg 104 = gap +16. Won Kentucky Derby at 23-1.

FADE

EV model says overbet chalk — market-implied probability significantly exceeds model win probability. Never singled. If gap ≥ 6 , kept as secondary in pairs/spreads. If gap < 6 , excluded entirely from ticket.

Rule: Rule change from backtest: original rule excluded FADE entirely. Shred the Gnar (gap +7, FADE) won R11 Oaks Day. New rule keeps gap ≥ 6 FADE horses as secondary.

Example: Knightsbridge: oFIG 112, ML 9-5, FADE $\rightarrow$ lost Churchill Downs S.

BOMB

Manual or podcast signal at a long price ($\geq 15-1$). Added to the leg's candidate pool regardless of the gap threshold. Overrides the gap-based filter. Critical on Oaks Day when two BOMB winners had gaps below any standard threshold.

Rule: Both Oaks Day BOMB winners (Corporate Power gap+3, Kathynmarissa gap+2 at 24-1) were below the ≥ 4 or ≥ 5 gap threshold. BOMB was the only signal that would have included them.

Example: Corporate Power: gap +3, BOMB tag $\rightarrow$ won Alysheba G2.

BOOKEND

Auto-detected sequence pattern where only the first and last legs have strong signals. Middle legs are automatically opened to 3 horses and labeled BOOKEND. Requires raising max cost setting to capture P5/P6 bookend sequences.

Rule: Only applies to sequences of 4+ legs. The BOOKEND label appears in the ticket header. Middle legs show BOOKEND $\times 3$ instead of the normal horse count.

Example: Oaks Day P6 R8–9–10–11–12–13: On Time Girl (R8, gap+10) and Always a Runner (R13, gap+7) were bookend anchors.

What the scanner does not do



Does not replace pace analysis

The scanner uses figure-based signals only. Fast fractions, pace mismatch, speed duels, and track bias are not modeled. Always layer in pace notes from the Race Day tab for stakes races — especially turf sprints and dirt routes with lone speed.



Does not work on fields with no OptixEQ data

Horses without oFIG figures (first-time starters, international shippers) get market-implied probability only — no gap signal, no EV edge. T O Elvis (Japan) won the Churchill Downs Stakes G1 at \$13.74 with no figure in the database. Flag any race with known international runners manually.



Uses morning line odds, not live tote

All model probabilities and EV calculations use morning line odds. If a horse drifts significantly from ML by post time, the EV call may change. Check live odds on the tote board before finalizing tickets — particularly for VALUE-flagged horses that may have been bet down.



Thin sample spans are less reliable

Sequence windows marked with a "thin sample" warning have fewer than 10 historical hits. The overlay ratio is real but may not reflect the true long-run distribution. Use thin-sample tickets as supplementary, not primary plays.

CD historical position overlays — reference

Wage r	Span	Hist. hits	Median payout	Overlay ratio	Consistency
P3	R2–3–4	47	\$238	1.25×	Moderate
P3	R5–6–7	48	\$261	1.22×	Moderate
P3	R8–9–10	38	\$315	1.54×	Moderate — best P3 span
P4	R4–5–6–7	13	\$1,127	2.56×	Lottery — top span
P5	R5–6–7–8–9	7	\$1,584	2.72×	Lottery (thin sample)
P5	R6–7–8–9–10	8	\$2,374	3.43×	Lottery (thin sample)

Will-pay median multipliers at CD: $P3 \rightarrow P4 = 4.6\times$ · $P4 \rightarrow P5 = 5.1\times$ · $P3 \rightarrow P5 = 23.8\times$